

Computer Networks – Complete Short Notes (Exam-Oriented)

1. Switching & Router Architecture

Switching moves packets from input ports to output ports using a switching fabric. Router architecture separates control plane (routing decisions) and data plane (fast forwarding).

- Input Port: line termination, link-layer processing, lookup/forwarding, queueing
- Switching Fabrics: memory, bus, crossbar/interconnection network
- Output Port: buffering, scheduling, link-layer transmission
- Control Plane (ms) vs Data Plane (ns)

2. Switch Fabric Types

- Memory-based: packet copied to memory; limited by memory bandwidth
- Bus-based: shared bus; contention limits throughput
- Crossbar/Interconnection: parallel paths; high performance, scalable

3. VLAN (Virtual LAN)

VLANs logically segment a switch into multiple broadcast domains, improving performance and security.

- Each VLAN = separate broadcast domain & subnet
- Reduces broadcast traffic and improves security
- Static VLANs (port-based) and Dynamic VLANs (VMPS)
- Inter-VLAN communication requires a router or Layer-3 switch

4. Spanning Tree Protocol (STP)

STP prevents Layer-2 loops by creating a loop-free logical topology.

- Issues: Layer-2 loops, broadcast storms, duplicate frames
- Uses BPDUs to elect Root Bridge (lowest Bridge ID)
- Select Root Ports, Designated Ports; block others
- Port cost based on link speed

5. IPv6 Overview

IPv6 was developed to overcome IPv4 exhaustion and improve scalability.

- 128-bit address space; hexadecimal notation
- No broadcast; uses multicast and anycast
- Stateless Address Auto Configuration (SLAAC)
- Simplified header; no checksum; extension headers

IPv6 Header & Extension Headers

- Fixed 40-byte base header
- Fields: Version, Traffic Class, Flow Label, Payload Length, Next Header, Hop Limit
- Extension headers: Hop-by-Hop, Routing, Fragment, AH, ESP

6. Router Configuration & ICMP

- Startup-config (NVRAM) vs Running-config (RAM)
- IOS stored in Flash; IFS provides unified file access
- ICMP: error reporting, diagnostics (ping, traceroute)
- Router discovery via ICMP messages

7. Optical Transmission & Multiplexing

- Uses light pulses through fiber (core, cladding, buffer)
- Advantages: high bandwidth, low attenuation, EMI immunity
- Fiber types: Multimode (step/graded), Single-mode
- Multiplexing: FDM, TDM, WDM, CDM

8. Network Delays & Loss

- Processing delay: router processing time
- Queuing delay: depends on congestion
- Transmission delay = $\text{packet length} / \text{link rate}$
- Propagation delay = $\text{distance} / \text{propagation speed}$

9. WAN Technologies

- Point-to-Point (leased lines)
- Circuit Switching (e.g., ISDN)
- Packet Switching (Frame Relay, ATM, X.25)
- Virtual Circuits: SVC and PVC

10. Data Link Control & ARQ Protocols

- Flow control: prevents receiver buffer overflow
- Error control via ARQ (retransmissions)
- Stop-and-Wait ARQ: simple, low efficiency
- Sliding Window: Go-Back-N, Selective Repeat (higher throughput)

End of Exam-Oriented Short Notes